

Add-on Occlusion: Building an Occlusion-capable Optical See-through Head-mounted Display with HoloLens 1

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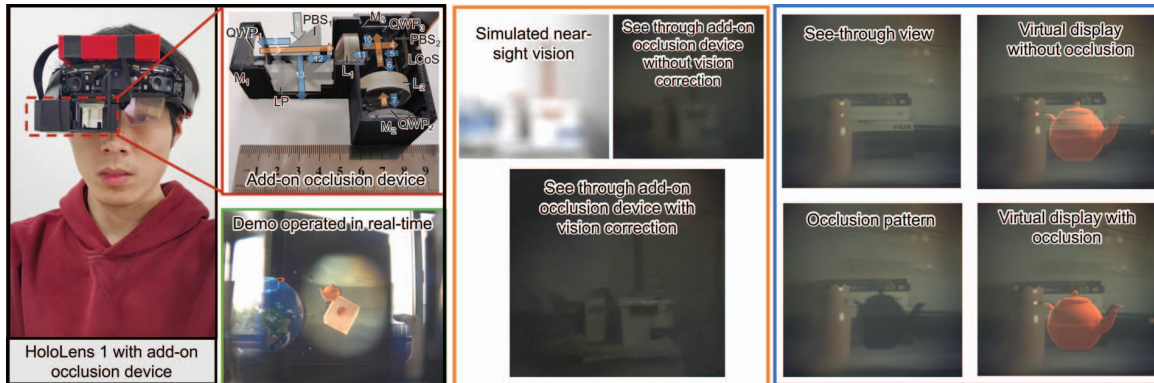


Figure 1: **Left:** Integrated device with HoloLens 1 (right eye's view). The add-on occlusion device is attached to the front surface of the waveguide combiner, and a driver board for operating the liquid crystal on silicon (LCoS device) is put in the red box. The optical architecture is shown in the red rectangle, which is highly compressed by reusing optical elements by modulating light polarization between the p-direction (orange arrows) and the s-direction (blue arrows) repeatedly. The prototype built in a 3D-printed housing achieves a volume of $8.5 \times 5.5 \times 3 \text{ cm}^3$. A demo built with Unity that enables object manipulation with occlusion capability is shown in the green rectangle. **Middle:** A camera is used to simulate users with near-sight, and a similarly blurred image is captured through the add-on occlusion device with default optical power. Then, the add-on occlusion device is set to correct the near-sight vision by changing the optical power, which is easily completed by shifting the lens position. **Right:** Mutual occlusion is demonstrated by the assembled system. The Spatial Awareness System of the Mixed Reality Tool Kit (MRTK) is used to build the real-world mesh, and the occlusion pattern is generated with the virtual object in real-time.

ABSTRACT

Hard-edge occlusion is demonstrated as a key feature for optical see-through head-mounted displays (OSTHMDs), while is currently not supported by any commodity product in the market. In this research demo, we provide HoloLens 1, which is one of the most widely used OSTHMDs, with a novel add-on occlusion device to build an occlusion-capable optical see-through head-mounted display (OC-OSTHMD). No modification is required on the optical system of HoloLens 1 since the proposed device works as an external module. A demo is built in Unity, proving the real-time virtual display with hard-edge occlusion is realized by the integrated system.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Hardware—Emerging technologies—Emerging optical and photonic technologies;

1 INTRODUCTION

The optical see-through head-mounted display (OSTHMD) is one of the most used platforms to implement augmented reality (AR)

applications nowadays. Although a variety of technologies are introduced into OSTHMDs that make the device more powerful in the virtual display, the perceived virtual image suffers the color blending issue caused by the background illumination [1]. Occlusion-capable optical see-through head-mounted displays (OC-OSTHMDs) that support hard-edge occlusion are demonstrated to tackle this problem by blocking the real scene on a pixel-by-pixel basis [2]. Virtual images displayed with hard-edge occlusion have been demonstrated to have higher contrast, a wider color gamut, and stronger perceptual cues [3]. However, a specific 4-f structure for OC-OSTHMDs causes a bulkier size and a smaller field of view (FOV) than common OSTHMDs. So far, there is no commodity product of OC-OSTHMDs available. Providing an easily accessed solution for OC-OSTHMDs is urgent and challenging.

Therefore, we propose add-on occlusion with this research demo. Different from existing OC-OSTHMDs that integrate a virtual display and the hard-edge occlusion function into an optical architecture, the add-on occlusion device is hung before the optical combiner of common OSTHMDs as an external module. The virtual display is implemented by the OSTHMD while hard-edge occlusion is realized by the add-on device, making the integrated device equivalently work as a typical OC-OSTHMD. Since the add-on occlusion device is simply fixed before the optical combiner, no modification is required on the optical system of the OSTHMD, which guarantees the compatibility of the proposed method with a wide range of OSTHMDs (e.g., HoloLens 1 & 2, Magic Leap 1, Nreal Light, etc.). Specific to this research demo, we pick HoloLens 1 as the device that integrated with the add-on occlusion device.

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2 OPTICAL DESIGN OF THE ADD-ON OCCLUSION DEVICE

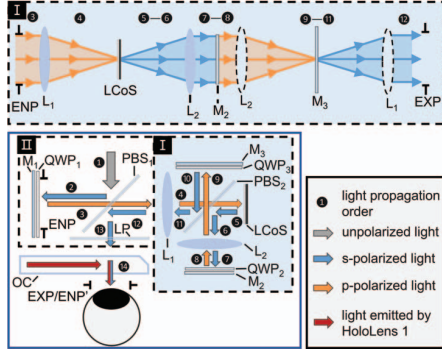


Figure 2: Optical design of the add-on occlusion device.

The optical design of the add-on occlusion device is shown in Fig. 2. Except for the optical combiner (OC) of HoloLens 1, the entire system in the solid blue rectangle consists of two parts, which are Part I and part II work for realizing hard-edge occlusion and making the device parallax-free, respectively. Unpolarized light (grey) from the real scene propagates the device by the sequence from (1) to (13), with the polarization direction switched between the s direction (blue) and the p direction (orange) repeatedly. The s-polarized light generated from the unpolarized light by a polarizing beam splitter (PBS₁) (1) is reflected toward a mirror M₁ (2). A quarter-wave plate (QWP₁) is attached to the front surface of the M₁. The combination of the M₁ and the QWP₁ transforms the incoming s-polarized light into the reflected p-polarized light, allowing it to pass through the PBS₁ and then reach a convex lens L₁ (3). Since the PBS₂ is placed to have the same polarization direction as the PBS₁, p-polarized light is transmitted and reaches the liquid crystal on silicon (LCoS) device (4). The non-occluded real scene is turned into s-polarized after being reflected backward on the LCoS surface (5) and guided to another convex lens L₂ by the PBS₂ (6). The M₂ and the QWP₂ are used to guide the light backward with the polarization being switched into the p direction (7)-(8). Then, the light passes the lens pair of L₂ and L₁ again with polarization changed by the M₃ and the QWP₃ (9)-(12). The folded light path from (3) to (12) is uncompressed as the equivalent structure in the upper dash rectangle. The reuse of L₁ and L₂ allows the device to form a typical 4-f system for realizing hard-edge occlusion in OC-OSTHMDs. Notice that the LCoS and the M₃ are placed at the back focal planes of L₁ and L₂, respectively. Finally, the processed real scene is guided to the user's eye (13). Since the add-on occlusion device is hung before the OC, the virtual display light (red) emitted by the projector of HoloLens 1 merges with the processed real scene light. The user putting his/her eye at the exit pupil (EXP) of the device perceives virtual images with occlusion enabled. Moreover, the entrance pupil (ENP) of Part I is conjugated into ENP' for overlapping with EXP, benefiting the add-on occlusion device with a parallax-free feature that avoids the viewpoint shift of the user.

A prototype built in a 3D-printed housing is shown in Fig. 1. L₂ is specially mounted into a lens holder that allows shifting along with the optical axis. Hence, the optical power of the prototype can be adjusted for users with near sight. As shown in the middle rectangle of Fig. 1, the real background at a distance of 3m is captured by a camera focusing at 0.5m, which leads to a blurred image as a near-sight user. Then, the add-on occlusion device is placed before the camera, the captured image turns to be sharp after L₂ is adjusted to correct the vision.

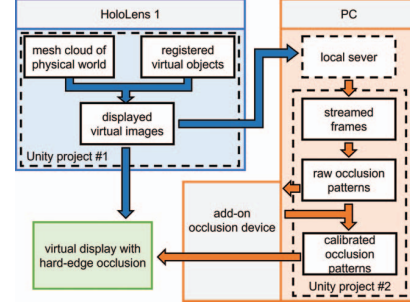


Figure 3: Overview of the integrated system of the add-on occlusion device and HoloLens 1.

3 INTEGRATED SYSTEM WITH HOLOLENS 1

Apart from the physical assembly of the add-on occlusion device and HoloLens 1, rendering occlusion patterns for virtual images is essential for operating the integrated system. Fig. 3 shows the pipeline for realizing the virtual display with hard-edge occlusion in the integrated device. Paired Unity projects are operated on HoloLens 1 (#1) and a PC connected with the add-on occlusion device (#2), respectively. The project #1 works for rendering virtual images with the registered virtual objects and captured mesh data of the physical world. Rendered virtual images are streamed and then stored in a local server held by the PC, where the project #2 runs for rendering occlusion patterns based on the streamed frames. Raw occlusion patterns are first rendered and sent to the add-on occlusion device, then calibrated to visually overlap the virtual images in the user's see-through view. The calibration data is pushed back to the shader in the project #2, guaranteeing the precise match between virtual images and occlusion patterns during the demo operation. As result, the integrated system provides users with a virtual display with hard-edge occlusion as typical OC-OSTHMDs.

Fig. 1 shows the real-time operation of the demo with the integrated device. We also demonstrate the crucial mutual occlusion feature supported by typical OC-OSTHMDs with the integrated device in the blue rectangle of Fig. 1. A red teapot is placed between a pink vase and a stack of books. A semi-transparent teapot is perceived when only the virtual display by HoloLens 1 is activated. Then, an occlusion pattern is shown to block the books in the background. As a result, the teapot shows higher visibility and better lighting effect with the occlusion pattern enabled.

4 CONCLUSION

The occlusion capability has been demonstrated to be a crucial feature for OSTHMDs, while a universal solution for the AR market is still under exploration. We propose add-on occlusion that allows off-the-shelf OSTHMDs to be occlusion-capable. The integrated device with HoloLens 1 is demonstrated in this research demo. We expect the proposed approach to provide a promising way to realize the universal application of hard-edge occlusion in OSTHMDs.

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